

Punctuated Institutional Equilibria: Stasis as Evolutionarily Stable Strategy, Basin Depth, and the Temporal Dynamics of Constitutional Lock-in

Ignacio Adrián Lerer

Independent Researcher

Buenos Aires, Argentina

adrian@lerer.com.ar

ORCID 0009-0007-6378-9749

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ABSTRACT

Evolutionary institutional theory explains *who* is selected (memes as replicators, institutions as vehicles) and *what pattern* characterizes long-run legal evolution (99.3% gradual, 0.7% punctuated). What remains unexplained is *when* and *why* institutional equilibria change. This paper addresses that temporal gap. I model institutional stasis as an Evolutionarily Stable Strategy (ESS) in the sense of Maynard Smith (1982): a configuration of legal norms, enforcement mechanisms, and professional networks such that no alternative configuration can invade when rare. Basin depth, the minimum perturbation required to escape an institutional ESS, is formalized as a function of the Constitutional Lock-in Index (CLI). Using replicator dynamics applied to four Latin American jurisdictions (Argentina CLI = 0.89, Spain CLI = 0.51, Brazil CLI = 0.40, Chile CLI = 0.24) and 60 reform episodes (1983-2025), I derive four falsifiable predictions. First, the magnitude of exogenous shock required for successful institutional punctuation correlates positively with pre-shock CLI ($\rho \geq 0.70$). Second, post-punctuation CLI drops by at least 0.20 within five years of successful regime transition. Third, failed reform attempts leave CLI unchanged (variation < 5%). Fourth, the Institutional Evolvability Index (IEI) predicts which jurisdictions are candidates for the next punctuation event. The framework reconciles Dawkins and Gould without distorting either: intermemetic selection (Level 1) assembles the ESS; emergent stability at the configuration level produces Gould's stasis; standard replicator dynamics on a shifting fitness landscape produce punctuation without requiring species-level selection as a separate causal force.

Keywords: punctuated equilibrium, evolutionarily stable strategy, replicator dynamics, constitutional lock-in, institutional stasis, basin of attraction, extended phenotype theory, evolutionary game theory, multilevel selection

I. Introduction: The Missing Temporal Dimension

Two recent contributions within the Extended Phenotype Theory (EPT) program have resolved distinct questions about legal evolution. The multilevel selection paper (Lerer, 2026a) addressed the vertical question: *who* is selected, at what level, through what causal mechanism? Its answer, building on Okasha's (2024) distinction between organizational and selection hierarchies, was that memes remain the replicators, institutions the vehicles, and that higher-level emergent effects operate as downward constraints rather than as independent selection units. The Constitutional Lock-in Index 3.0, formalized through lattice theory and the Birkhoff polytope, captured these constraints mathematically.

The Back Slope paper (Lerer, 2025) addressed the longitudinal question: *what pattern* characterizes legal evolution across deep time? Mapping 6-7 million years of institutional precursors, from primate reciprocity tracking to constitutional specialization, it documented a ratio of 139 parts gradual accumulation to 1 part revolutionary transition. The 99.3%/0.7% split validated both Dawkins's emphasis on incremental selection and Gould and Eldredge's (1972) emphasis on punctuated transitions as complementary descriptions of a single process.

Neither paper answered the temporal question that comparative institutional analysis most urgently requires: *when* will an institutional equilibrium change, and *why* does change, when it arrives, come as a rupture rather than as gradual adjustment? This is not a theoretical luxury. The difference between a jurisdiction in deep stasis (Argentina's labor regime, stable since 1957) and a jurisdiction approaching a bifurcation point (Brazil's labor code in 2016, one year before the Temer reform) is invisible to static indices. Both exhibit high resistance to reform at any given moment. The distinction lies in the dynamic properties of the equilibrium: its depth, its sensitivity to perturbation, and the proximity of exogenous pressures to the invasion threshold.

This paper proposes a formal framework for that distinction. I model institutional stasis as an Evolutionarily Stable Strategy (ESS) in the technical sense introduced by Maynard Smith and Price (1973) and elaborated by Maynard Smith (1982). An institutional ESS is a configuration of legal norms, enforcement mechanisms, and professional networks such that no feasible alternative can increase in frequency when initially rare. Basin depth, defined as the minimum perturbation to payoff parameters required to destabilize the ESS, is derived as a function of the Constitutional

Lock-in Index. Punctuation occurs when exogenous shocks push payoff parameters beyond the invasion threshold, triggering rapid convergence to a new ESS through standard replicator dynamics.

The contribution is threefold. First, it supplies the temporal dimension absent from the hierarchical framework of Lerer (2026a). That paper explained why institutions resist change (downward constraints from accumulated phenotypic machinery). This paper explains when that resistance breaks. Second, it generates falsifiable predictions that neither static CLI measurement nor qualitative historical analysis can produce: specifically, that shock magnitude required for punctuation is a measurable function of pre-shock CLI. Third, it reconciles Dawkins and Gould without the theoretical distortions that plagued earlier attempts. Gould’s stasis is not evidence for species selection; it is the observable consequence of an ESS assembled through intermemetic selection. Punctuation is not a special macroevolutionary force; it is what replicator dynamics look like when fitness landscapes shift abruptly.

II. Institutional Stasis as Evolutionarily Stable Strategy

A. The ESS Concept Translated to Institutions

Maynard Smith (1982) defined an ESS as a strategy such that, if adopted by a population, no mutant strategy can invade under the influence of natural selection. The formal condition is straightforward: a strategy S is an ESS if, for all alternative strategies $T \neq S$, either (i) the payoff of S against S exceeds the payoff of T against S , or (ii) these payoffs are equal but the payoff of S against T exceeds the payoff of T against T . The first condition means the incumbent outperforms any challenger in the normal environment; the second means that if the challenger achieves parity under normal conditions, the incumbent still wins in head-to-head encounters.

Translating this to institutional analysis requires specifying what counts as a “strategy,” what constitutes the “population,” and how “payoff” is measured. I adopt the following definitions, consistent with the EPT framework developed in prior work. A *strategy* is a configuration of legal norms governing a specific institutional domain (e.g., labor law, tax law, commercial regulation). It includes not only the formal rules but the interpretive doctrines, enforcement practices, and

professional training that operationalize them. A *population* is the set of institutional actors within a jurisdiction who interact under these norms: judges, legislators, practitioners, regulated parties, and enforcement agencies. *Payoff* is measured as institutional fitness in the EPT sense: the capacity of a normative configuration to persist, replicate across jurisdictions, and resist displacement by alternatives.

Under these definitions, Argentina's labor regime illustrates an institutional ESS with unusual clarity. The configuration assembled since the 1957 constitutionalization of Article 14 bis consists of: (a) expansive constitutional text with vague protective language ("estabilidad del empleado público," "retribución justa," "participación en las ganancias"); (b) 187 specialized labor courts applying pro-worker interpretive doctrines; (c) a dense network of labor law professionals trained in university programs organized around the protective principle; (d) powerful union organizations with constitutional standing and institutional resources; and (e) administrative agencies enforcing labor regulations with bureaucratic inertia resistant to political direction. Each component reinforces the others. Labor courts apply doctrines taught in universities; universities train practitioners who become judges; unions litigate in courts to expand protective interpretations; administrative agencies enforce the expanded interpretations. The configuration's payoff, measured as persistence and resistance to invasion, exceeds that of any alternative. Twenty-three reform attempts between 1991 and 2025 failed to displace it.

Why can't a reform strategy invade? The ESS condition provides the answer. Consider a hypothetical "flexibilization" strategy T that proposes to simplify dismissal procedures, reduce mandatory benefits, and decentralize collective bargaining. When T is rare (a single reform bill introduced in a legislature dominated by the incumbent configuration), its payoff against S is negative: labor courts strike down the reform, unions mobilize against it, practitioners trained under the incumbent framework challenge it doctrinally, and administrative agencies delay or obstruct implementation. The payoff of S against a rare T is positive: the incumbent configuration absorbs the challenge and may even strengthen itself through the judicial precedents generated in defeating the reform. The ESS condition is satisfied.

B. Frequency Dependence and Institutional Reinforcement

The institutional ESS exhibits a specific form of frequency dependence that merits explicit formalization. Let x represent the proportion of institutional actors operating under the incumbent configuration S , and $(1 - x)$ the proportion attempting to operate under the alternative T . The payoff functions can be written as:

$$W(S) = a + b \cdot x + c \cdot x^2$$

$$W(T) = d + e \cdot (1 - x) + f \cdot (1 - x)^2$$

where a is the baseline payoff of the incumbent in isolation, b captures linear network effects (more actors under S increase each actor's payoff through coordination benefits), c captures nonlinear reinforcement (the payoff advantage accelerates as the configuration becomes more entrenched), and d, e, f are the corresponding parameters for the alternative. The critical feature is that institutional configurations typically exhibit $b > 0$ and $c > 0$ simultaneously: network effects and nonlinear reinforcement operate in the same direction. This is because institutional components are complementary, not substitutable. A labor court's value to a union increases when more courts apply the same doctrine; a university program's value to graduates increases when more courts demand the training it provides. The complementarity generates positive feedback loops absent in many biological contexts, where frequency-dependent selection often operates through diminishing returns (the payoff of a strategy declines as it becomes more common because resources are depleted).

The implication is that institutional ESS tend to be exceptionally stable. Where biological ESS are often maintained by a balance of opposing frequency-dependent pressures (hawk-dove dynamics, for instance), institutional ESS are maintained by self-reinforcing complementarity. The basin of attraction surrounding an institutional ESS is not merely deep; it deepens over time as the institutional infrastructure accumulates. This is the mechanism that generates the hysteretic effect captured by the Institutional Hysteresis Rate (IHR): even if the norms that created the institutional infrastructure are formally repealed, the infrastructure persists and continues to stabilize the incumbent configuration.

III. Basin Depth and the Constitutional Lock-in Index

A. Defining Basin Depth

In dynamical systems theory, a basin of attraction is the set of initial conditions from which a system converges to a particular equilibrium. Basin depth, informally, is the minimum perturbation required to push the system out of one basin and into another. For institutional analysis, I define basin depth (δ) as the minimum magnitude of exogenous payoff shock required to violate the ESS condition, enabling an alternative configuration to invade.

Formally, let $\Delta W = W(S) - W(T)$ evaluated at $x = 1$ (the incumbent is universal). The ESS condition requires $\Delta W > 0$. An exogenous shock of magnitude σ modifies payoff parameters, reducing ΔW by some amount $g(\sigma)$. Basin depth δ is the value of σ such that $\Delta W - g(\delta) = 0$. In plain language: basin depth is the shock just sufficient to make the incumbent and the challenger equally fit, opening the door to invasion.

B. CLI as Proxy for Basin Depth

The Constitutional Lock-in Index, validated with an AUC of 0.97 across 60 international cases, measures the structural resistance of an institutional configuration to reform. Its five components, treaty hierarchy (TV), judicial activism (JA), textual vagueness (TH), precedent weight (PW), and amendment difficulty (AD), capture precisely the mechanisms that widen the payoff gap ΔW between incumbent and challenger. High textual vagueness (TH) allows judicial expansion of the incumbent's scope, increasing $W(S)$. High judicial activism (JA) means courts actively defend the incumbent against challengers. High precedent weight (PW) increases the coordination cost of switching to an alternative. High treaty hierarchy (TV) adds external constraints protecting the incumbent. High amendment difficulty (AD) raises the procedural cost of formal change.

Each CLI component thus maps onto a specific mechanism that increases ΔW and, correspondingly, basin depth. The relationship can be expressed as:

$$\delta = k \cdot \text{CLI} / (1 - \text{CLI})$$

where k is a scaling constant and the hyperbolic form captures the nonlinear relationship between lock-in and basin depth: as CLI approaches 1.0, basin depth increases hyperbolically. This is empirically plausible: the difference in reform difficulty between CLI 0.24 (Chile) and CLI 0.40 (Brazil) is far smaller than the difference between CLI 0.40 (Brazil) and CLI 0.89 (Argentina). The formula predicts that Argentina's basin depth is approximately 12 times Chile's: $(0.89/0.11)$

$/ (0.24/0.76) = 8.09/0.32 \approx 25.6$. Even adjusting for the crudeness of the estimate, the order-of-magnitude difference is consistent with observed reform outcomes: Chile has successfully reformed its labor code six times since 1990; Argentina has succeeded zero times in 23 attempts.

C. The Deepening Mechanism: Why Basins Are Not Static

A distinctive feature of institutional basins, absent in most biological and physical analogs, is that they deepen endogenously over time. The mechanism is the accumulation of phenotypic machinery described in the EPT framework. Every year that Argentina's labor regime persists without successful challenge, new institutional infrastructure is built: additional judicial precedents reinforcing the protective principle, new generations of lawyers trained exclusively in the incumbent doctrine, expanded administrative capacity oriented toward enforcement of existing norms. Each increment of infrastructure widens the payoff gap ΔW and increases basin depth. The IHR captures the irreversible component of this accumulation: infrastructure that persists even after formal norms are repealed.

This endogenous deepening creates a temporal asymmetry that is critical for understanding institutional dynamics. Early in an ESS's history, basins are shallow: the configuration has not yet accumulated sufficient infrastructure, and reform is feasible with moderate shocks. As the ESS matures, basins deepen: infrastructure accumulates, complementarities solidify, and progressively larger shocks are required for punctuation. Argentina's CLI increased from approximately 0.65 at the time of Article 14 bis's enactment in 1957 to 0.89 in 2025, reflecting six decades of phenotypic machinery accumulation. The early reform window, roughly 1957-1970, when the institutional infrastructure was still incomplete, was the period of lowest basin depth. That window was not exploited; by the time comprehensive reform was attempted (beginning with Menem's labor flexibility program in the 1990s), the basin had deepened beyond what any politically feasible shock could overcome.

IV. The Mechanics of Punctuation

A. Exogenous Shocks and the Invasion Threshold

If stasis is the norm, punctuation requires explanation. Within the ESS framework, the explanation is precise: punctuation occurs when an exogenous shock σ exceeds basin depth δ , enabling an alternative configuration to invade. The shock must be exogenous to the institutional system itself; endogenous pressures (internal contradictions, accumulating inefficiencies) are absorbed by the ESS through the same adaptive mechanisms that resist reform.

Three categories of shock are empirically relevant for institutional punctuation. *Economic crises* alter payoff structures by making the incumbent configuration visibly costly: hyperinflation, sovereign default, or prolonged recession reduces the fitness of existing institutional arrangements and increases the relative attractiveness of alternatives. *Regime discontinuities* reset the institutional landscape by removing veto players: military coups, revolutions, or democratic transitions eliminate or weaken the actors who enforce the incumbent ESS. *External pressures* modify the competitive environment: IMF conditionality, trade agreements, or geopolitical realignment change the payoff structure that sustains the incumbent.

Brazil's 2017 labor reform (Law 13.467) illustrates the mechanics. Brazil's labor regime, anchored in the 1943 Consolidação das Leis do Trabalho (CLT), had a pre-reform CLI of approximately 0.40: resistant to reform but not impenetrably locked in. The exogenous shock was a combined economic-political crisis: the deepest recession in Brazilian history (GDP contracted 3.5% in 2015 and 3.3% in 2016), followed by the political upheaval of Dilma Rousseff's impeachment in August 2016. The shock exceeded basin depth. The Temer administration, operating under exceptional political conditions (an impeachment-installed government with nothing to lose electorally), pushed through comprehensive reform in less than six months. The transition was abrupt, exactly as replicator dynamics predict: once the invasion threshold was crossed, the new configuration displaced the old rapidly rather than through gradual convergence.

Contrast Argentina during the same period. Argentina experienced comparable economic distress (inflation exceeding 40% in 2016, recession, fiscal crisis), and the Macri administration (2015-2019) attempted labor reform with substantial political capital. The reform failed. The difference is explained by basin depth: Argentina's CLI of 0.89 meant that the required shock was approximately an order of magnitude larger than Brazil's. The 2016 Argentine economic crisis, severe by any standard, was insufficient to exceed the invasion threshold of a basin that had been deepening since 1957.

B. Speed of Transition: Why Punctuation Is Rapid

A puzzle that has attracted attention in both biological (Gould, 2002) and institutional (Pierson, 2004) literatures is why transitions, when they occur, are rapid rather than gradual. The ESS framework provides a precise answer. Once the invasion threshold is crossed (once $\sigma > \delta$), the dynamics are governed by the replicator equation:

$$dx/dt = x \cdot (1 - x) \cdot [W(S) - W(T)]$$

When $W(T) > W(S)$ (the challenger outperforms the incumbent under shocked conditions), x decreases toward zero: the incumbent loses frequency. The rate of change is proportional to $x \cdot (1 - x)$, which is maximized at $x = 0.5$ and approaches zero at the boundaries. The dynamics thus have a characteristic S-shaped trajectory: slow initial invasion (when the challenger is rare), rapid displacement (as the challenger approaches 50% frequency and the logistics function peaks), and slow final consolidation (as the incumbent approaches extinction). The middle phase, where the bulk of institutional displacement occurs, is compressed into a brief period. This is the formal mechanism behind the empirical observation that successful reforms, when they happen, happen fast.

Chile's experience confirms this pattern. The six successful labor reforms since 1990 each followed a similar trajectory: extended political preparation (analogous to the slow initial phase), rapid legislative passage and implementation (the fast middle phase), and gradual institutional consolidation as courts, practitioners, and agencies adapted to the new regime (the slow final phase). The entire displacement phase, from bill introduction to functional implementation, typically lasted 12-18 months. What preceded it, the accumulation of political conditions making reform feasible, took years.

V. Empirical Application: Four Jurisdictions

A. Comparative Predictions

The framework generates specific, testable predictions for the four jurisdictions whose CLI values have been empirically calibrated. The logic is straightforward: higher CLI implies deeper basin,

which implies larger shock required for punctuation, which implies longer expected duration of stasis and lower probability of reform success at any given moment.

Jurisdiction	CLI	Basin Depth (relative)	Reforms Attempted	Reforms Succeeded	Predicted Shock Threshold
Argentina	0.89	25.6x	23	0 (0%)	Regime discontinuity
Spain	0.51	3.3x	13	9 (69%)	Moderate economic crisis
Brazil	0.40	2.1x	14	6 (43%)	Severe econ. + political
Chile	0.24	1.0x (base)	10	8 (80%)	Normal political cycle

Table 1. Comparative Basin Depth and Reform Outcomes across Four Latin American Jurisdictions (1983-2025). Basin depth is calculated as $CLI/(1-CLI)$, normalized to Chile as the baseline. Reform data from Lerer (2025b, 2026a).

B. Argentina: Deep Stasis and the Absent Punctuation

Argentina's 0/23 reform success rate over 34 years (1991-2025) is the empirical anchor for the deep-basin hypothesis. The nine attempts under Menem (1991-1999), four under De la Rúa (1999-2001), two under Kirchner governments (2003-2015), and eight under Macri and subsequent administrations all failed through the same mechanism: the institutional ESS absorbed each challenge through judicial invalidation, union mobilization, bureaucratic obstruction, or some combination of the three. Even the 2001 economic catastrophe, the most severe crisis in Argentine modern history (GDP contraction of 10.9%, unemployment exceeding 25%, sovereign default), did not produce labor reform. This is a powerful test of the basin depth hypothesis: if a shock of that magnitude could not displace the ESS, the basin is extraordinarily deep.

The framework predicts that Argentina's labor ESS will persist until a shock exceeding the current basin depth occurs. Given the hyperbolic relationship between CLI and basin depth, the required shock is not merely large but qualitatively different from ordinary economic crises. The model suggests that only a regime discontinuity, a constitutional convention, a revolutionary change in political structure, or an equivalent disruption that removes or neutralizes the veto players embedded in the ESS, could exceed the threshold. Incremental reform is not merely difficult; it is, under current conditions, dynamically impossible.

C. Chile: Shallow Basin and Repeated Punctuation

Chile's 8/10 reform success rate over 35 years (1990-2025) demonstrates the opposite dynamic. With a CLI of 0.24, basin depth is shallow: normal political cycles generate sufficient payoff perturbation for alternative configurations to invade. Chile's labor code has been successfully reformed in both directions, increasing and decreasing worker protections, confirming that low CLI measures bidirectional reform capacity rather than ideological orientation. The 2001 reform expanded collective bargaining rights; the 2016 reform further strengthened them; earlier reforms under military-era codes had reduced them. Each transition followed the S-shaped replicator dynamic: slow preparation, rapid displacement, gradual consolidation.

The 2022 constitutional convention provides a particularly informative case. The convention drafted a replacement constitution with expansive labor provisions resembling Argentina's Article 14 bis, potentially increasing Chile's CLI from 0.24 to an estimated 0.65-0.70. Chilean voters rejected the draft 62-38% in the September 2022 referendum. Post-referendum polling by the Centro de Estudios Públicos (CEP) indicated that 47% of voters cited fears of excessive labor rigidity as a reason for rejection. In ESS terms, the Chilean polity rejected a deeper basin. The population, operating under a shallow-basin regime that permitted repeated adaptation, chose to preserve that adaptive capacity rather than lock in a deeper equilibrium. This is a remarkable case of institutional self-awareness: a jurisdiction consciously avoiding the trap that Argentina fell into in 1957.

D. Brazil: Intermediate Basin and the Temer Punctuation

Brazil occupies the intermediate position that makes the framework's predictions most testable. With a CLI of 0.40, Brazil's labor basin was deep enough to resist routine reform pressure but shallow enough to be punctured by a sufficiently severe shock. The 2015-2016 economic crisis (cumulative GDP contraction of approximately 7%) combined with the Rousseff impeachment created shock conditions exceeding basin depth. The Temer reform of 2017, which modified over 100 provisions of the CLT, was the resulting punctuation event.

The post-reform trajectory confirms the model's predictions about consolidation dynamics. The new configuration has not yet fully stabilized: judicial challenges continue, implementation is

uneven across Brazilian states, and some provisions have been modified by subsequent legislation. The replicator dynamics predict this slow consolidation phase (the right tail of the S-curve), and they predict that the new ESS will eventually stabilize at a lower CLI, approximately 0.30-0.35, reflecting the reduced institutional infrastructure of the reformed system. Whether this prediction holds will require longitudinal CLI measurement over the coming decade.

VI. Reconciling Dawkins and Gould through EGT Dynamics

The Dawkins-Gould debate over units of selection has generated more heat than light, partly because the interlocutors were often talking past each other. Dawkins (1982, 1986) insisted that the gene (or, in the cultural domain, the meme) is the replicator and that all interesting evolutionary change reduces to selection among replicator variants. Gould (2002) insisted that species exhibit genuine individuality, that stasis is a real pattern requiring explanation at the species level, and that macroevolutionary trends reflect species selection rather than extrapolated microevolution.

The ESS framework applied to institutions resolves this debate by showing that both positions are correct, at different analytical levels, in a way that requires no theoretical compromise from either side.

Dawkins is correct about assembly. The institutional ESS is assembled through intermemetic selection at Level 1. Individual norms, doctrines, and practices compete; those with higher fitness (greater capacity to replicate across jurisdictions, persist over time, and resist displacement) spread at the expense of alternatives. The protective principle in Argentine labor law, the doctrine of *in dubio pro operario*, the concept of *estabilidad*: each of these memes was selected through competition with alternatives. No special “species-level” selection was required to assemble the configuration. Dawkins’s gene-centered (meme-centered) framework fully explains how the ESS came to exist.

Gould is correct about stasis. Once assembled, the ESS exhibits stability that is a property of the configuration as a whole, not reducible to the properties of its component memes. A single norm within the Argentine labor ESS, considered in isolation, might be displaced by a superior alternative. But the norm is not in isolation; it is embedded in a complementary network where displacement of one component is resisted by all others. The stasis is emergent: it arises from the

interaction structure, not from any individual component's intrinsic stability. This is precisely Gould's point about species individuality. The difference is that, in the institutional case, the mechanism generating the emergent stability (complementarity among memes within a configuration) is fully specified in terms of the lower-level units. Gould's empirical observation is preserved; his inference to species selection as a *causal* mechanism is replaced by the ESS mechanism, which is causal but operates through interactions among the lower-level units that Dawkins champions.

Punctuation, in this framework, is what replicator dynamics look like when fitness landscapes shift abruptly. It requires no special macroevolutionary force. When an exogenous shock exceeds basin depth, the ESS condition is violated, and standard replicator dynamics drive the system toward a new equilibrium. The transition is rapid because replicator dynamics near the midpoint of the S-curve are inherently fast, not because some mysterious species-level selection mechanism operates during crises. Gould's empirical observation, that transitions are abrupt and concentrated rather than gradual, is fully explained. His theoretical interpretation, that a distinct macroevolutionary mechanism is at work, is unnecessary.

The reconciliation can be stated in a single sentence: Dawkins explains how institutional ESS are assembled; Gould describes what institutional ESS look like from the outside; evolutionary game theory explains when they change.

VII. Falsifiable Predictions

The framework generates four predictions that are testable with existing or readily constructible data. Following the EPT program's commitment to Popperian falsifiability, I specify for each prediction the conditions under which it would be refuted.

H1: Shock-CLI Correlation. The magnitude of exogenous shock required for successful institutional punctuation correlates positively with pre-shock CLI. Specifically, across the 60-case dataset, Spearman's ρ between pre-reform CLI and the severity of the triggering crisis (measured as a composite of GDP contraction, political disruption, and external pressure) should exceed 0.70 for successful reforms. *Falsification:* $\rho < 0.50$, or a successful major reform in a jurisdiction with $\text{CLI} > 0.80$ without an identifiable severe exogenous shock.

H2: Post-Punctuation CLI Drop. Successful institutional punctuation reduces CLI by at least 0.20 within five years of the reform event. The mechanism is the destruction of phenotypic machinery: specialized courts are reformed, curricula are updated, administrative agencies are restructured. *Falsification:* A jurisdiction experiences successful comprehensive reform but CLI remains within 0.10 of its pre-reform value five years later.

H3: Failed Reform Invariance. Failed reform attempts leave CLI unchanged (variation $< 5\%$ of the pre-attempt value). The ESS absorbs the challenge without structural modification. *Falsification:* A clearly failed reform (judicially invalidated, legislatively repealed, or abandoned during implementation) produces CLI change $> 10\%$.

H4: IEI as Punctuation Predictor. The Institutional Evolvability Index, particularly its V (variation generation) and M (modularity) components, predicts which jurisdictions are candidates for the next punctuation event. Jurisdictions with $IEI > 0.50$ that experience exogenous shocks exceeding their (lower) basin depth should punctuate; jurisdictions with $IEI < 0.30$ should resist even severe shocks. *Falsification:* A jurisdiction with $IEI < 0.20$ experiences successful comprehensive reform, or a jurisdiction with $IEI > 0.60$ fails to reform despite a shock exceeding the predicted threshold.

VIII. Discussion

A. Comparison with Existing Frameworks

Three existing frameworks address institutional persistence: North's (1990) path dependence, Pierson's (2004) increasing returns, and Mahoney's (2000) self-reinforcing sequences. The ESS framework shares features with each but differs in important respects.

North's path dependence explains persistence through switching costs: once an institutional path is chosen, the cost of changing paths increases over time. This is correct but underspecified. It does not explain why some institutional paths generate higher switching costs than others, nor does it generate quantitative predictions about when switching will occur. The ESS framework adds precision: switching costs are a function of basin depth, which is a function of CLI, which is

measurable. The prediction is not merely “reform is hard” but “reform requires a shock of magnitude $\sigma > k \cdot \text{CLI}/(1-\text{CLI})$.”

Pierson’s increasing returns captures the self-reinforcing dynamics that the ESS framework formalizes through positive frequency dependence. Pierson’s account, however, is descriptive rather than formal: it identifies the pattern (increasing returns generate lock-in) without specifying the equilibrium conditions (ESS), the invasion dynamics (replicator equation), or the breaking conditions (exogenous shock exceeding basin depth). The ESS framework translates Pierson’s qualitative insights into a formal apparatus that generates testable predictions.

Mahoney’s self-reinforcing sequences add historical specificity: he identifies the critical juncture at which a path is chosen, the reactive sequences that follow, and the conditions under which path reversal becomes possible. The ESS framework complements Mahoney by formalizing what constitutes a “critical juncture”: it is the moment when an exogenous shock exceeds basin depth, enabling a new ESS to displace the incumbent. The advantage of the formal specification is predictive: rather than identifying critical junctures retrospectively (as Mahoney does), the framework aspires to anticipate them by monitoring CLI, IEI, and exogenous shock indicators.

B. Limitations

Four limitations require candid acknowledgment. First, the basin depth formula ($\delta = k \cdot \text{CLI}/(1-\text{CLI})$) is a theoretically motivated estimation, not an empirically calibrated measurement. The hyperbolic form is plausible (reform difficulty increases nonlinearly with CLI) but the scaling constant k and the functional form itself await rigorous empirical validation. The formula should be treated as a first approximation, useful for generating testable predictions but not yet as a confirmed relationship.

Second, the measurement of exogenous shock magnitude is inherently multidimensional and difficult to standardize. GDP contraction, political disruption, and external pressure do not reduce to a single scalar without contested weighting choices. The correlation predicted in H1 depends on a defensible composite measure that does not yet exist in standardized form.

Third, the framework assumes that stasis is the product of ESS dynamics rather than mere inertia or ignorance. It is possible that some institutional persistence reflects neither complementarity nor

frequency-dependent reinforcement but simply the absence of anyone with sufficient motivation to push for change. Distinguishing ESS-maintained stasis from inertia-maintained stasis requires more granular empirical work than this paper provides.

Fourth, the four-jurisdiction sample, while theoretically motivated and empirically rich, is small. Extending the analysis to additional jurisdictions (Colombia, Peru, Mexico, the post-socialist transitions of Eastern Europe) would substantially strengthen the empirical foundation. The existing 60-case dataset provides a starting point; future work should expand it.

IX. Conclusion

Institutional stasis is not the absence of evolution; it is evolution's most stable product. The ESS framework developed in this paper provides the formal apparatus for understanding why institutions resist change (the equilibrium is self-reinforcing), when they change (exogenous shocks exceed basin depth), and how fast the change occurs (replicator dynamics produce S-shaped transitions with a rapid displacement phase).

The framework fills the temporal gap between two prior contributions. Where Lerer (2026a) specified the hierarchical architecture (who is selected, at what level, through what causal chain), this paper specifies the dynamics (when equilibria shift, at what speed, under what conditions). Where Lerer (2025) documented the long-run pattern (99.3% gradual, 0.7% punctuated), this paper explains the mechanism generating that pattern: ESS-maintained stasis produces the 99.3%; exogenous shocks exceeding basin depth produce the 0.7%.

The practical implication is sobering for jurisdictions trapped in deep institutional basins. Argentina's 0/23 reform failure rate is not a streak of bad luck, political incompetence, or insufficient will. It is the predictable consequence of a configuration whose basin depth exceeds any politically feasible shock. Reform in such jurisdictions requires either patience (waiting for an exogenous shock of extraordinary magnitude) or ambition (engineering a regime discontinuity that removes the veto players embedded in the ESS). Gradual reform is not merely unlikely; in the formal sense specified by the model, it is dynamically impossible when basin depth exceeds the feasible shock range.

The research agenda opened by this framework points in two directions. The first is the formal estimation of basin depth through longitudinal CLI measurement, enabling the framework to move from theoretical prediction to empirical calibration. The second, developed in a companion paper currently in preparation, is the construction of an institutional fossil record: systematic CLI, IHR, and IEI measurement across jurisdictions over decades, enabling future researchers to observe, in compressed form, the patterns of stasis and punctuation that current observers can only conjecture from within.

IA Use Disclaimer: *The theoretical framework, hypotheses, and analytical argument in this paper are entirely the author's. The formal expressions (payoff functions, basin depth formula, replicator equation) were developed by the author with verification assistance from artificial intelligence tools, which were also used for literature review and drafting support. The author reviewed, revised, and takes full responsibility for all content, including all mathematical formulations.*

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